

## THE ROLE OF INDUSTRIAL ENGINEERS IN INDUSRTY

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### Abstract

This paper explains and discusses the role of Industrial Engineer (IE) in industry. Authors have observed that there is a confusion on what an IE actually does in industry? The authors also have observed that students in industrial engineering departments have some difficulties in explaining their profession and role in their future professional life as IEs. This papers aims to give a clear understanding about the industrial engineering and its role to the people who have interest in this profession.

### INTRODUCTION

The definition of industrial engineering is given as follows by the three industrial engineering departments in Turkey.

Middle East Technical University: "Industrial Engineering deals with the solution of the problems that occur during the phases of planning, design, implementation and development in organisations, which produce goods or services. Especially the efficient use of scarce production resources and the development of methodologies and techniques to increase quality in production are its incidental functions. Industrial engineering uses the knowledge and skills of mathematics, natural and social sciences, engineering analysis and design to achieve this aim." (<http://www.ie.metu.edu.tr/dep.html#general>)

Bilkent University: "Industrial Engineering is concerned with how best to organise people, information, money, and materials to produce and distribute services and products. It draws upon specialised knowledge and skills in the mathematical, physical, and social sciences together with the principles and methods of engineering analysis and design to specify, predict and evaluate the results to be obtained from such systems." (<http://www.ie.bilkent.edu.tr/generalinfo.html>)

The definition of Gaziantep University is as follows: "Industrial Engineering has a multi-disciplinary nature. It combines several subjects from various engineering and social science fields to develop solutions to the problems (mainly related to engineering management) of the industry. The program in the Industrial Engineering mainly covers the disciplines of production engineering, operations management, operations research, ergonomics, CAD/CAM, automation, information technology, systems simulation, analysis

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*and design. The interdisciplinary nature of the program prepares the students for a decision making position within the planning and production functions of public and private enterprises and operations."*

It can be seen from the above definitions that the industrial engineering has an *interdisciplinary nature*. The reason for this can be explained by the diversity of decisions that the industrial engineers are expected to give like, organising (or giving decisions on organising) people-money-materials in order to produce and distribute high quality services and products. IE has always been integration engineering. While the traditional role of the IEs has been to integrate human, material, equipment, and financial resources into productive systems, the current emphasis is on the integration of computers, information, and technology to operate and control complex systems (Yaman, 1997).

It can also be seen from the above definitions that the main concentration of IE is on "production". Therefore, it is possible to relate the goals and role of the nonindustrial engineering to the main goals of the "production". This may give a better understating of the industrial engineering and some insights on its education. The work done by Köksal and Eğıtman (1997) can also be a good reference on IE education. In the following sections of this paper, the industrial engineering profession and its education will be underlined from this perspective.

### **INDUSTRIAL ENGINEERING PROFESSION**

The topmost goals of the production are known as **cost**, **quality** and **time** (see Figure 1). Endless ongoing effort is spent over the centuries to achieve these goals. The industrial engineering was emerged as a profession to generate solutions to the problems of the production in order to satisfy its goals.

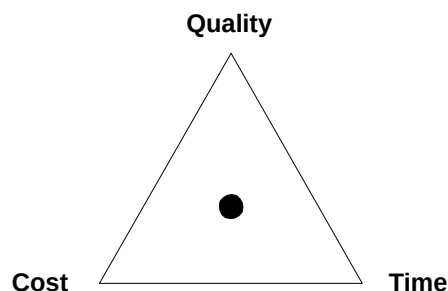


Figure 1. The main goals of the production

An IE should balance the main goals of the production (cost - quality - time) in any company by considering its **limited resources**, **working environment** (generally has dynamic nature) and **opportunities** in order to have a *competitive company*. In achieving this goal, IE uses many **diverse techniques** and **philosophies**. Therefore, a good IE should have a solid understanding of these objectives, techniques and philosophies. Modern production management accepts profit, internal-external customer satisfaction and responsiveness as the main assets of a company. Figure 2 illustrates graphically what an IE should try to achieve in a continuously changing production environment to maximise profit, internal-external customer satisfaction and quality. As it can be seen from Figure 2, there is direct relationship between the main goals of the production and the assets of a company (i.e. minimising cost maximises the profit, maximising the quality maximises the customer satisfaction, minimising the time maximises the responsiveness). It can also be seen from Figure 2 that, there is a triple trade-off between the goals and assets. As stated above, the IE plays an important role in resolving these trade-offs to maximise the company assets. This should be also considered as one of the main concentration point in industrial engineering education.

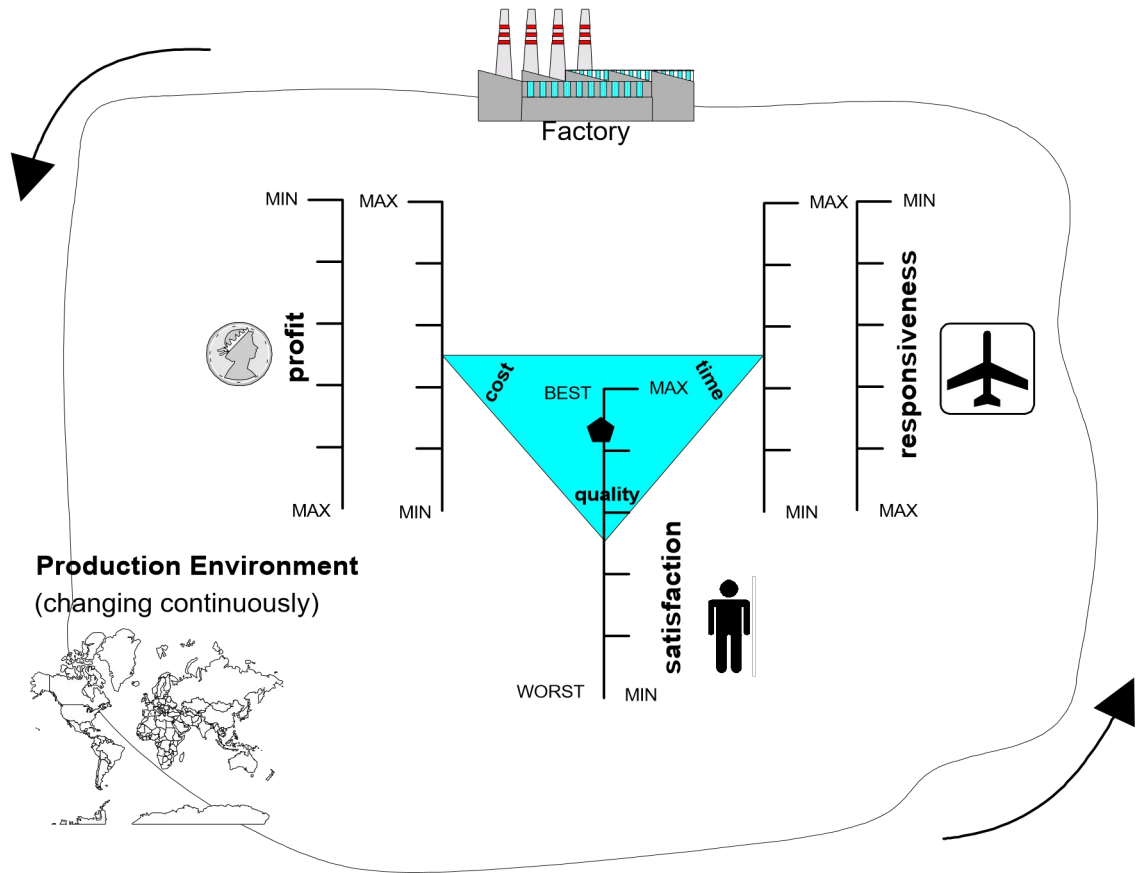


Figure 2. The role and goals of IE in a changing production environment

The main objective is of course to maximise all of the company assets as depicted in Figure 3. This is the ultimate goal of companies. Figure 4 shows the worst possible solution where the company survival is not easy.

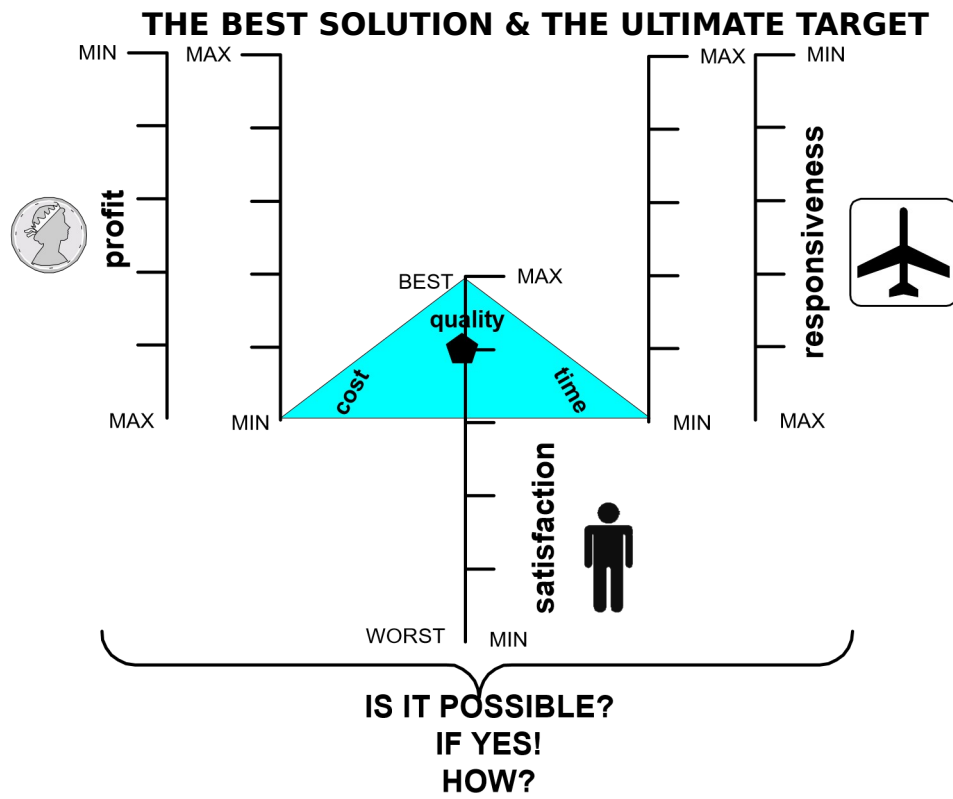


Figure 3. The best performing company

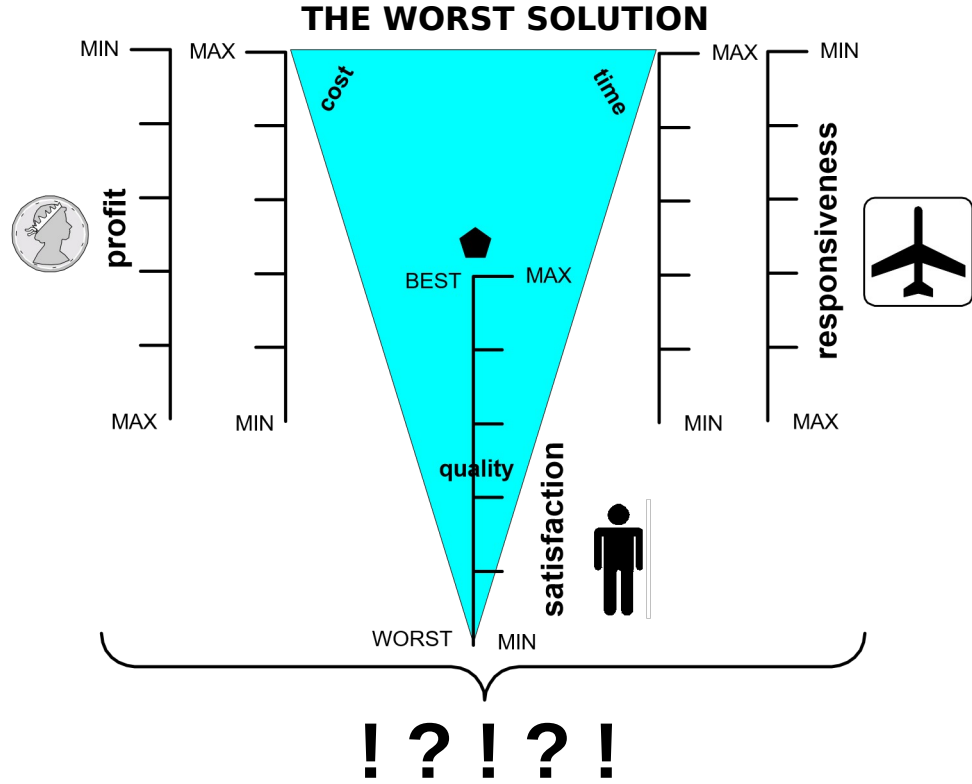


Figure 4. The worst performing company

On one hand, there are three basic interactions, like cost increase, quality improvement and time reduction, between the goals of production. Careful examinations of these interactions explain what should be done in order to simultaneously maximise company assets. It also gives insights on the industrial engineering education and the courses thought. Figure 5 depicts the first interaction (i.e. cost reduction) and its reflection to the courses thought in the Industrial Engineering Department of the University of Gaziantep (see Appendix Industrial Engineering Curriculum of the University of Gaziantep).

As it can be seen from Figure 5, reducing the cost does not mean to stop spending money. In contrary, the important issue is to make proper investment, HOW? Nobody makes investment to reduce the profit! Moreover, making only investment will generally not be sufficient to maximise the company assets, human effort is essential! Therefore, an IE should be able to resolve all these issues.

Figure 6 illustrates the second interaction (i.e. quality improvement). As it can be seen from this figure improving quality can reduce cost and time. Human effort is still essential and will speed-up quality improvement.

Figure 7 shows the third interaction (i.e. time reduction). As it can be seen from this figure reducing time can improve the quality and the reduce cost. Human effort still has big importance as shown Figure 7.

## INTERACTIONS-1

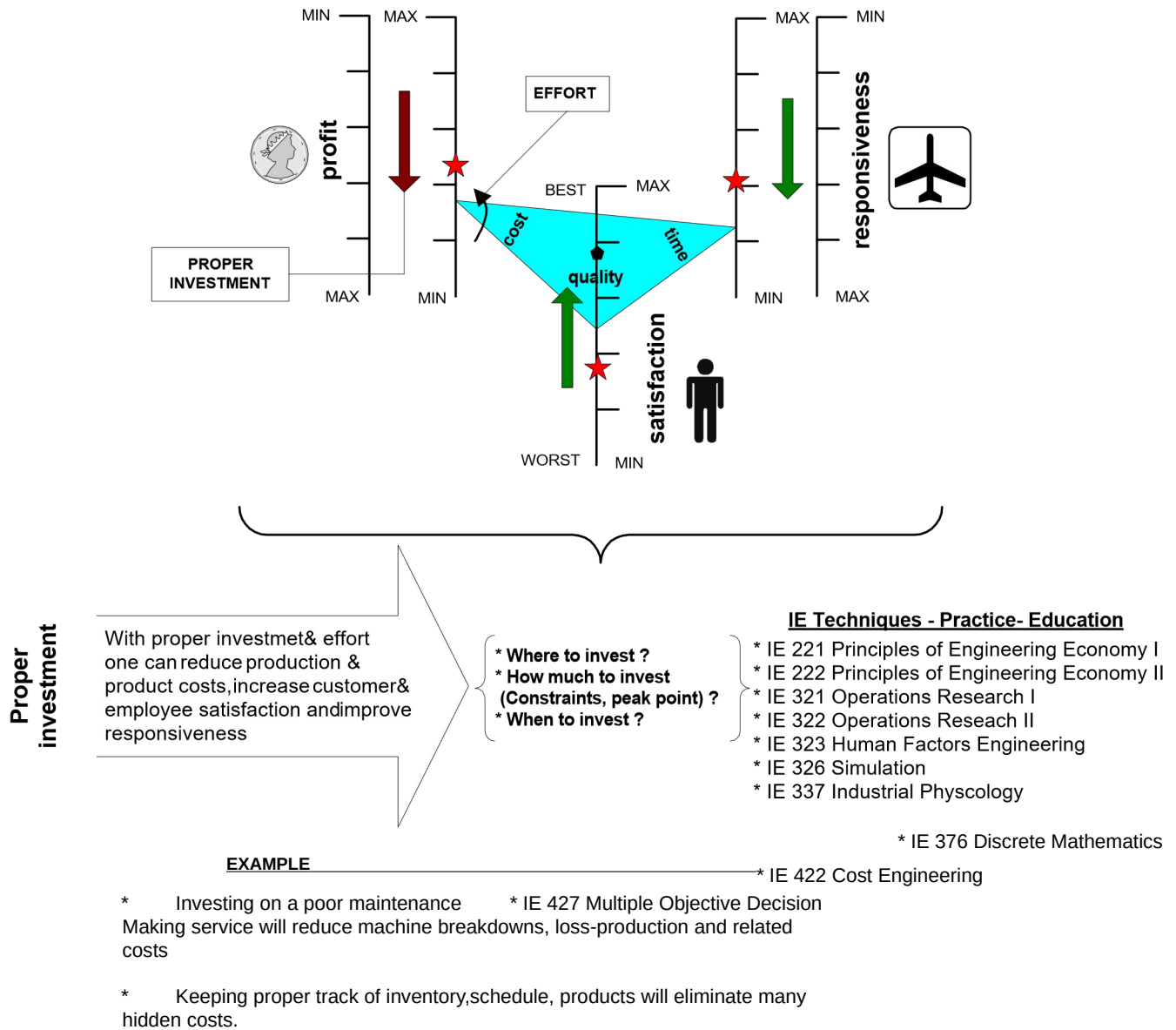


Figure 5. Interaction 1: Cost reduction



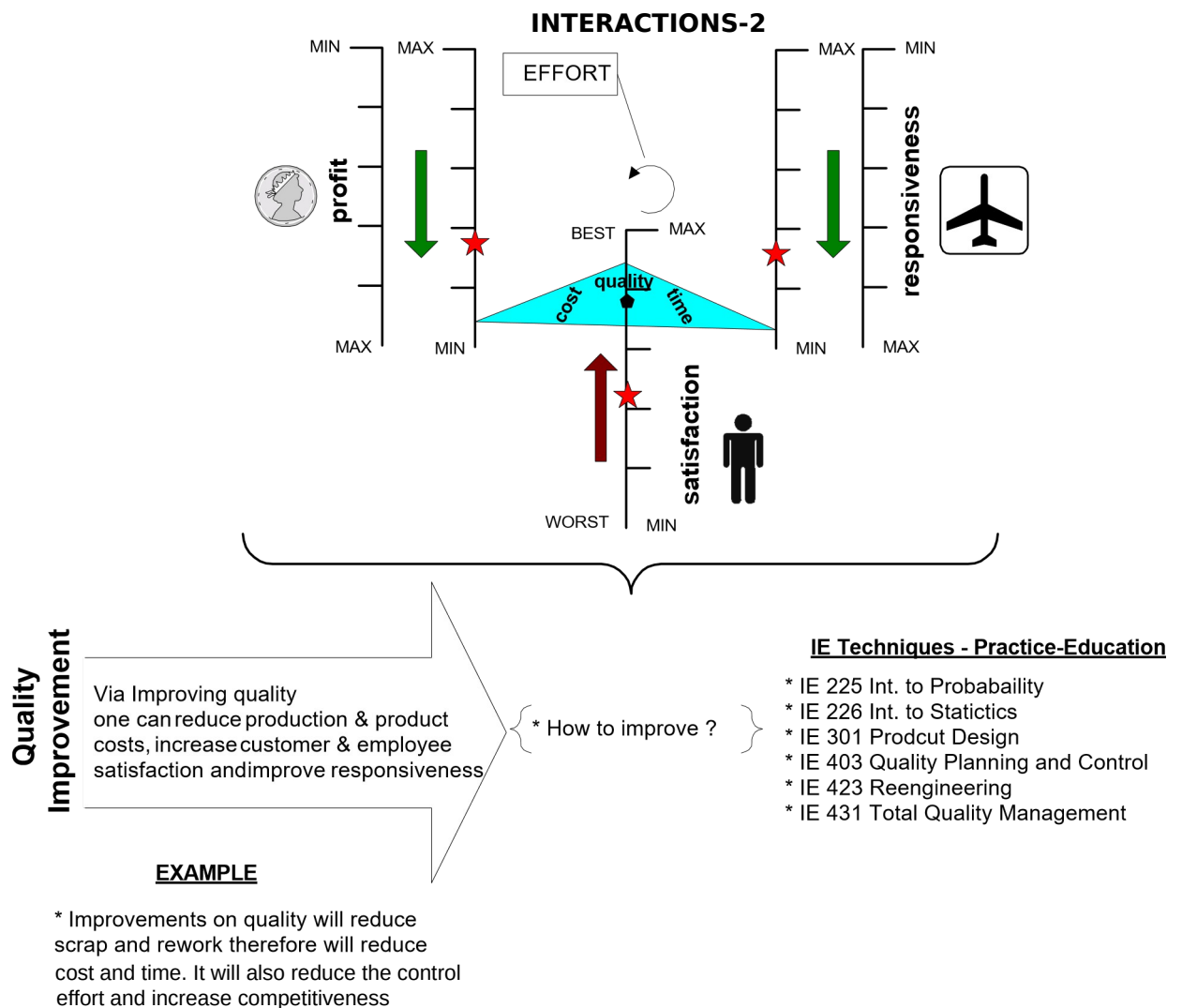


Figure 6. Interaction 2: Quality improvement

### INTERACTIONS-3

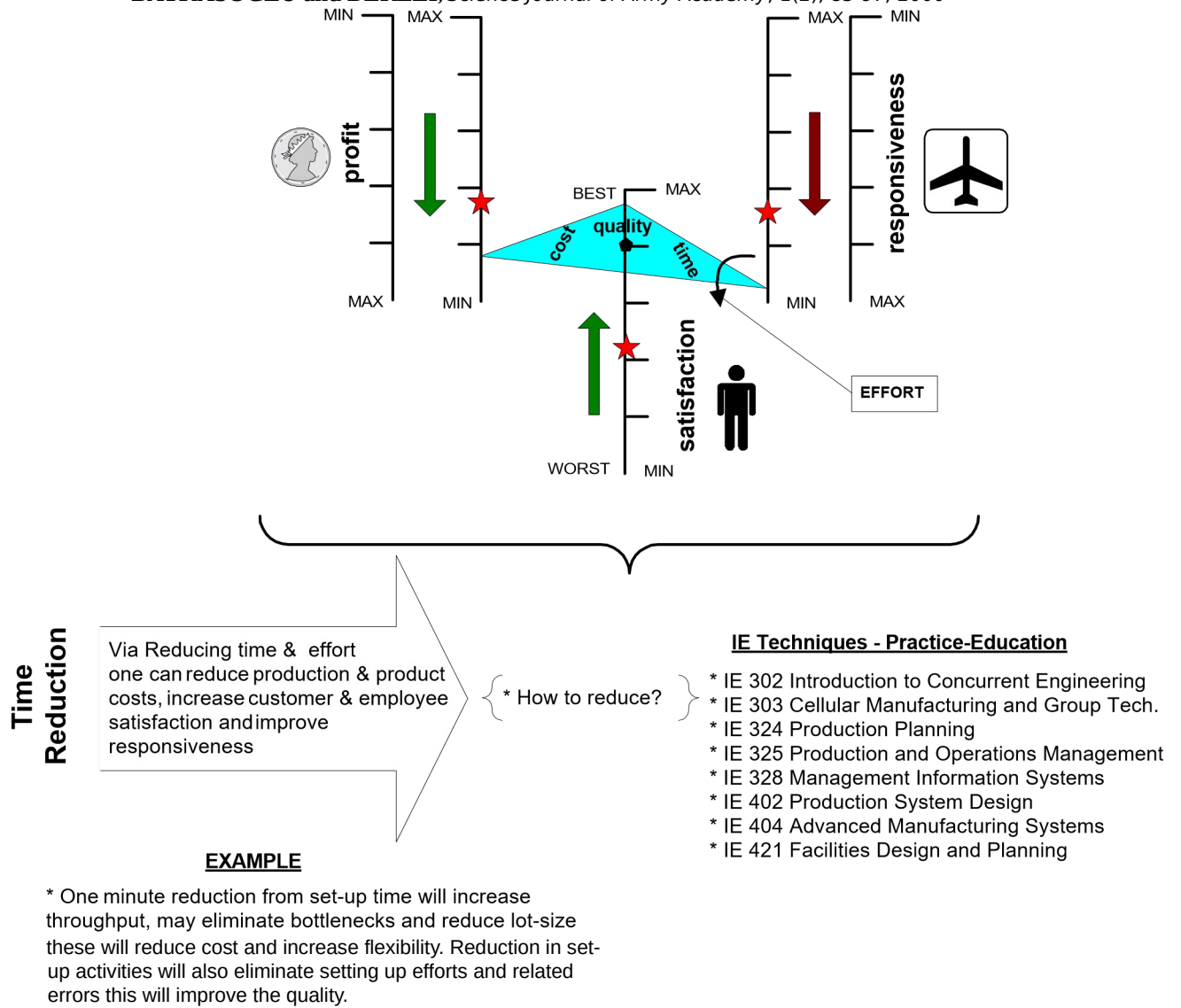


Figure 7. Interaction 3: Time reduction

## CONCLUSIONS

As it can be understood from the above self-explanatory figures and discussions, the industrial engineers play an important role in the industry for achieving the main goals of the production. They mainly try to manage and integrate human resources and technology to maximise assets of companies. Keeping the ultimate objectives of production in mind, their education should be planned accordingly and dynamically.

## REFERENCES

KÖKSAL, G., and EĞİTMAN, A.

1997 "Planning and Design of Industrial Engineering Education Quality", 22<sup>nd</sup> International Conference on Computers and Industrial Engineering, December 20-22, 1997 Cairo, Egypt pp.240-243.

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1997 "Expectations from an Industrial Engineer for Today's Industry", 22<sup>nd</sup> International Conference on Computers and Industrial Engineering, December 20-22, 1997 Cairo, Egypt pp.232-235.

<http://www.ie.metu.edu.tr/dep.html#general>

<http://www.ie.bilkent.edu.tr/generalinfo.html>

## APPENDIX

### UNDERGRADUATE CURRICULUM FOR INDUSTRIAL ENGINEERING

#### FIRST YEAR

##### First Semester

|      |     |                                        |                  |     |                    |                |     |
|------|-----|----------------------------------------|------------------|-----|--------------------|----------------|-----|
| MATH | 151 | Calculus I                             | ( 4 – 0 ) 4 EP   | 105 | General Physics I  | ( 4 – 0 ) 4 FE | 103 |
|      |     | General Chemistry                      | ( 3 – 1 ) 4 Leng | 101 | Freshman English 1 | ( 4 – 0 ) 4 IE | 141 |
|      |     | Computer Programming I                 | ( 2 – 0 ) 2      |     |                    |                |     |
| IE   | 161 | Introduction to Industrial Engineering |                  |     |                    | ( 2 – 0 ) 2    |     |
| TURK | 100 | Turkish                                |                  |     |                    | ( 2 – 0 ) 2    |     |

##### Second Semester

|      |     |                                   |                |     |                        |                  |     |
|------|-----|-----------------------------------|----------------|-----|------------------------|------------------|-----|
| MATH | 152 | Calculus II                       | ( 4 – 0 ) 4 EP | 106 | General Physics II     | ( 4 – 0 ) 4 Leng | 102 |
|      |     | Freshman English II               | ( 4 – 0 ) 4 ME | 101 | Engineering Graphics I | ( 2 – 2 ) 4      |     |
| IE   | 172 | Statics and Strength of Materials |                |     |                        | ( 4 – 0 ) 4      |     |
| TURK | 200 | Turkish                           |                |     |                        | ( 2 – 0 ) 2      |     |

|      |     |                                    |                        |
|------|-----|------------------------------------|------------------------|
| BESR | 100 | Physical Education                 | ( 1 – 1 ) non – credit |
| IE   | 199 | Industrial Engineering Orientation | non – credit           |

## SECOND YEAR

### Third Semester

|      |                |                                                                |             |                           |             |                             |             |
|------|----------------|----------------------------------------------------------------|-------------|---------------------------|-------------|-----------------------------|-------------|
| MATH | 255            | Linear Algebra                                                 | ( 3 – 0 ) 3 | ME                        | 215         | Engineering Materials I     | ( 3 – 1 ) 4 |
| IE   | 201            | Introduction to Manufacturing Processes                        | ( 2 – 2 ) 4 | IE                        | 205         | Introduction to Probability |             |
|      | ( 3 – 0 ) 3    | IE                                                             | 221         | Principles of Economics I | ( 3 – 0 ) 3 | IE                          | 241         |
|      | Programming II | ( 2 – 1 ) 3                                                    |             |                           |             |                             | Computer    |
| HIST | 100            | Atatürk's Principles and the History of the Turkish Renovation | ( 2 – 0 ) 2 |                           |             |                             |             |

### Fourth Semester

|      |     |                                                                |             |    |     |                          |              |
|------|-----|----------------------------------------------------------------|-------------|----|-----|--------------------------|--------------|
| MATH | 256 | Differential Equations                                         | ( 3 – 0 ) 3 | ME | 208 | Engineering Mechanics II | ( 3 – 1 ) 4  |
| IE   | 202 | Computer Graphics and Geometric Modelling for IEs              | ( 2 – 0 ) 2 |    |     |                          |              |
| IE   | 222 | Principles of Economics II                                     | ( 3 – 0 ) 3 |    |     |                          |              |
| IE   | 226 | Introduction to Statistics                                     | ( 3 – 0 ) 3 |    |     |                          |              |
| IE   | 228 | Work Study                                                     | ( 2 – 1 ) 3 |    |     |                          |              |
| IE   | 242 | Data Structures and Algorithms                                 | ( 2 – 0 ) 2 |    |     |                          |              |
| HIST | 200 | Atatürk's Principles and the History of the Turkish Renovation | ( 2 – 0 ) 2 |    |     |                          |              |
| IE   | 299 | Summer Practice                                                |             |    |     |                          | non – credit |

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**THIRD YEAR****Fifth Semester**

|    |     |                                      |                |     |                           |
|----|-----|--------------------------------------|----------------|-----|---------------------------|
| IE | 321 | Operations Research I                | ( 3 – 1 ) 4 IE | 323 | Human Factors Engineering |
|    |     |                                      | ( 2 – 2 ) 4    |     |                           |
| IE | 325 | Production and Operations Management | ( 3 – 0 ) 3 IE | 327 | Management for Engineers  |
|    |     | Elective Course                      |                |     | ( 3 – 0 ) 3               |
|    |     | Elective Course                      |                |     | ( 3 – 0 ) 3               |

**Sixth Semester**

|    |     |                                |  |  |              |
|----|-----|--------------------------------|--|--|--------------|
| IE | 322 | Operations Research II         |  |  | ( 3 – 1 ) 4  |
| IE | 324 | Production Planning            |  |  | ( 3 – 0 ) 3  |
| IE | 326 | Simulation                     |  |  | ( 2 – 2 ) 4  |
| IE | 328 | Management Information Systems |  |  | ( 3 – 0 ) 3  |
|    |     | Elective Course                |  |  | ( 3 – 0 ) 3  |
|    |     | Elective Course                |  |  | ( 3 – 0 ) 3  |
| IE | 399 | Summer Practice                |  |  | non – credit |

**FOURTH YEAR****Seventh Semester**

|    |     |                              |                |                                       |                                |
|----|-----|------------------------------|----------------|---------------------------------------|--------------------------------|
| IE | 403 | Quality Planning and Control | ( 4 – 0 ) 4 IE | 421                                   | Facilities Design and Planning |
|    |     | ( 2 – 2 ) 4 IE               | 425            | Engineering Economy and Cost Analysis | ( 3 – 0 ) 3                    |
|    |     | Elective Course              |                |                                       | ( 3 – 0 ) 3                    |
|    |     | Elective Course              |                |                                       | ( 3 – 0 ) 3                    |
|    |     | Elective Course              |                |                                       | ( 3 – 0 ) 3                    |

**Eight Semester**

|    |     |                          |                |     |                                |
|----|-----|--------------------------|----------------|-----|--------------------------------|
| IE | 402 | Production System Design | ( 4 – 0 ) 4 IE | 404 | Advanced Manufacturing Systems |
|    |     | ( 3 – 0 ) 3              |                |     |                                |
| IE | 499 | Graduation Project       | ( 0 – 4 ) 4    |     | Elective Course                |
|    |     | Elective Course          |                |     | ( 3 – 0 ) 3                    |
|    |     | Elective Course          |                |     | ( 3 – 0 ) 3                    |

**ELECTIVE COURSES**

|    |     |                                                     |    |     |             |
|----|-----|-----------------------------------------------------|----|-----|-------------|
|    |     |                                                     |    |     | ( 3 – 0 ) 3 |
| IE | 301 | Product Design                                      |    |     | ( 3 – 0 ) 3 |
| IE | 302 | Introduction to Concurrent Engineering              |    |     | ( 3 – 0 ) 3 |
| IE | 303 | Introduction to Computer Integrated Manufacturing I |    |     | ( 3 – 0 ) 3 |
| IE | 304 | Cellular Manufacturing and Group Technology         |    |     | ( 3 – 0 ) 3 |
| IE | 329 | Productivity Management                             | IE | 331 | ( 3 – 0 ) 3 |
|    |     | Dynamic Programming                                 | IE | 332 | Modern      |
|    |     | Heuristics                                          |    |     | ( 3 – 0 ) 3 |
| IE | 333 | Graph Theory and Networks                           | IE | 334 | Numerical   |
|    |     | Analysis                                            |    |     | ( 3 – 0 ) 3 |
| IE | 335 | Introduction to Forecasting                         | IE | 336 | ( 3 – 0 ) 3 |
|    |     | Production Scheduling                               |    |     | ( 3 – 0 ) 3 |
| IE | 337 | Industrial Psychology                               |    |     | ( 3 – 0 ) 3 |
| IE | 338 | Logistics                                           |    |     | ( 3 – 0 ) 3 |
| IE | 339 | Introduction to Cost Accounting                     |    |     | ( 3 – 0 ) 3 |
| IE | 344 | Int. to Object Oriented Design and Programming      |    |     |             |
| IE | 347 | Computer Applications                               |    |     | ( 3 – 0 ) 3 |
| IE | 370 | Non-linear Programming Applications                 |    |     | ( 3 – 0 ) 3 |
| IE | 376 | Discrete Mathematics                                |    |     | ( 3 – 0 ) 3 |

|        |                                                                  |        |                                                       |
|--------|------------------------------------------------------------------|--------|-------------------------------------------------------|
| IE 401 | Computer Aided Design and Manufacturing ( 3 – 0 ) 3              | IE 405 | Computer Numerical Control Programming ( 3 – 0 ) 3    |
| IE 406 | Introduction to Robotics ( 3 – 0 ) 3                             |        |                                                       |
| IE 407 | Introduction to Computer Integrated Manufacturing II ( 3 – 0 ) 3 |        |                                                       |
| IE 408 | Lean Production ( 3 – 0 ) 3                                      | IE 409 | Responsive Manufacturing ( 3 – 0 ) 3                  |
| IE 410 | Mechatronics ( 3 – 0 ) 3                                         | IE 422 | Cost Engineering ( 3 – 0 ) 3                          |
| IE 423 | Reengineering ( 3 – 0 ) 3                                        |        |                                                       |
| IE 424 | Material Requirements Planning ( 3 – 0 ) 3                       | IE 426 | Business Process Modelling and Simulation ( 3 – 0 ) 3 |
| IE 427 | Multiple Objective Decision Making ( 3 – 0 ) 3                   | IE 428 | Introduction to Enterprise Modelling ( 3 – 0 ) 3      |
| IE 429 | Work Analysis and Evaluation ( 3 – 0 ) 3                         | IE 431 | Total Quality Management ( 3 – 0 ) 3                  |
| IE 441 | Advanced Computer Algorithms ( 3 – 0 ) 3                         | IE 461 | System Dynamics and Control ( 3 – 0 ) 3               |
| IE 462 | Control System Design ( 3 – 0 ) 3                                |        |                                                       |
| IE 463 | Introduction to Control System Technology ( 3 – 0 ) 3            |        |                                                       |
| IE 464 | Expert Systems ( 3 – 0 ) 3                                       |        |                                                       |
| IE 465 | Artificial Intelligence in Engineering ( 3 – 0 ) 3               |        |                                                       |
| IE 471 | Energy Management ( 3 – 0 ) 3                                    | IE 472 | Textile Engineering ( 3 – 0 ) 3                       |
| IE 473 | Thermodynamics and Heat Power ( 3 – 0 ) 3                        |        |                                                       |
| IE 474 | Marketing ( 3 – 0 ) 3                                            |        |                                                       |
| IE 475 | Information Technology ( 3 – 0 ) 3                               | IE 476 | Distance Learning ( 3 – 0 ) 3                         |
| IE 477 | Implementation of Advanced Manufacturing Systems ( 3 – 0 ) 3     |        |                                                       |
| IE 478 | Case Studies in Industrial Engineering ( 3 – 0 ) 3               |        |                                                       |